Changes of chamber generators and binary translation groups

Human Compression Project

Abstract

We study changes of distinguished chamber generators using subgroup intersections, coset representatives and the objects retained by the operation. Endpoint conjugation has an exact intersection criterion; its geometric realization is a component of a graph's bipartite double cover. We also give a common algebraic description of the binary translations that occur in halving generalized cubes and in polyhedra with equal adjacent parameters. Edge differences in a connected graph present its augmentation module; an explicit covariance identity then identifies groups generated by these translations with semidirect products. For the cyclic displacement relations used by Li and Zhang, this gives mutually inverse homomorphisms and proves the universal presentation without comparing finite group orders. The arguments isolate the graph generation and presentation recognition steps so that geometric applications can use the same statements with their own distinguished generators.

1 Introduction

Many constructions by involutions have a translation subgroup whose elements are even subsets of a finite set. The fact that the subset is even explains its size, but it does not by itself prove that the proposed generators produce every translation, or that an abstract presentation has no additional kernel. These two questions have a common answer in terms of differences of vertices in a connected graph.

We distinguish the underlying module from its chosen presentation. Paths express every difference from a base vertex as a sum of edge differences; cycle relations are exactly the ambiguity in this expression. Once a group permutes the vertices, the corresponding conjugation formula determines the whole semidirect product. This gives an explicit inverse map in the cyclic application, rather than a surjection followed by an equality of orders.

The binary-coordinate construction and its applications to generalized cubes are attributed to Danzer and to Montero–Ivić Weiss. The commutativity of the cyclic displacements is an expanded argument from Li–Zhang. The formulation below reorganizes these existing ingredients into statements that can be used by both applications. No historical priority is asserted for graph cycle spaces, augmentation modules or semidirect products. The geometric statements use two distinct hypotheses. Intersections of subgroups generated by subsets of markers identify flag stabilizers; a common point for pairwise incident cosets supplies flag completion. The two hypotheses are kept separate in the recognition theorem. For an operation retaining vertices or edges, its action kernel is computed on those actual retained objects, rather than inferred from an abstract presentation. This distinction is needed by Petrial operations, whose natural action need not be faithful.

The graph-halving construction, the lattice interpretation of nondegeneracy and the index-two criterion are attributed to the halving literature, including [PT24]. The endpoint support criterion, exact parity compatibility and retained-object formulation reuse the project's previously reviewed complete arguments. We include their proofs here: another topic's possession of them does not reduce this topic's common-theory cost.

2 Coset geometries and the objects retained by an operation

A geometry has a finite set of types; its flags are sets of pairwise incident objects of distinct types, every flag extends to a chamber, and its residues are the induced geometries on objects incident with a fixed flag. It is thin when every panel has exactly two completions and residually connected when every residue of rank at least two has connected incidence graph. Regularity means chamber transitivity of the full type-preserving automorphism group. No finiteness of the object set is implicit.

A marking of a group G in the geometric sections is a finite indexed generating family of nonidentity involutions $(r_i)_{i\in I}$. Write $G_J = \langle r_j : j \in J\rangle$. The intersection property is

$$G_J \cap G_K = G_{J\cap K} \qquad (J, K \subseteq I). \tag{C1}$$

The maximal parabolics are $P_i = G_{I\setminus\{i\}}$; their right cosets are incident when they intersect. Typed coset Helly means that every pairwise intersecting family with distinct types has a common element. UC.Geometry shows that IP and typed Helly give exactly the regular thin residually connected coset geometry, with G as its full type-preserving group. IP alone is not sufficient for that conclusion.

Proposition UC.Geometry (coset recognition and faithful chamber action). Suppose the marking of G consists of nonidentity involutions, has IP, and has typed coset Helly for its maximal parabolics. The coset incidence system is then a thin, residually connected geometry, and G is its full type-preserving automorphism group, acting regularly on chambers. More generally, the full type-preserving automorphism group of any thin, residually connected geometry of finite rank acts freely on chambers. A chamber-transitive subgroup is consequently the full group.

Proof. In the coset system, a common coset point extends every flag to a chamber by choosing the coset of each missing type through that point. Right translation sends every such flag to its base flag. For a base flag of type set S, IP gives stabilizer $G_{I\setminus S}$. In particular the full chamber stabilizer is 1, while a panel stabilizer is a group generated by one nonidentity involution and has order two. There are therefore exactly two chambers through each panel.

Consider a residue with remaining type set T, where $|T| \geq 2$. Its stabilizer is G_T, and its maximal object stabilizers are $G_{T\setminus\{i\}}$, $i \in T$. Every marker indexed by T lies in at least one of these proper parabolics. A word in those subgroups supplies a walk between their translated base objects in the residue incidence graph; all objects are translates of base objects and all base objects lie in a common flag. The residue is therefore connected.

For the general assertion, first observe that every residually connected finite-rank geometry has a connected chamber graph. This follows by induction on rank: chambers through a fixed object are connected through galleries in its residue, and a path in the incidence graph, extending each consecutive incident pair to a chamber, connects arbitrary chambers through these subgraphs. The rank-one case holds because all chambers share the empty panel. In a thin geometry an automorphism fixing a chamber fixes its unique neighbor across each panel. It therefore fixes every chamber along every gallery, and hence every object. The chamber action is free. If a subgroup is chamber-transitive, any full automorphism agrees on a base chamber with one subgroup element; their quotient fixes that chamber and is the identity. Apply this to the coset geometry to obtain the stated full automorphism group. □

Lemma C.Gallery (the marked stabilizers). In a thin regular residually connected geometry, the full group acts freely on chambers. At a base chamber let r_i be the unique group element carrying it across its i-panel. Then r_i is an involution, the r_i generate the full group, and the stabilizer of a base subflag F is

$$G_F = \langle r_i : i \notin t(F)\rangle. \tag{C2}$$

Proof. UC.Geometry proves chamber-graph connectedness and freeness of the chamber action. The square of r_i fixes the base chamber, so r_i has order two. A chamber gallery expresses every chamber image by successive panel changes, which gives generation. Apply the same gallery argument in the residue at F. Only the missing types can change, so its chamber-transitive stabilizer is generated by their r_i. Conversely those r_i fix F. This proves (C2). □

Lemma UC.Helly (anchored criterion). A finite indexed subgroup family $(P_i)_{i\in I}$ has typed coset Helly if and only if the following condition holds. For every $i \in I$, every $J \subseteq I \setminus \{i\}$ with $|J| \geq 2$, and any selected $j_0 \in J$, each $g \in P_{j_0}$ satisfying $g \in P_i P_j$ for all $j \in J$ admits $k \in \bigcap_{j\in J} P_j$ with $P_i g = P_i k$.

Proof. The cosets $P_i g$ and P_j intersect precisely when $g \in P_i P_j$. Thus Helly gives a common element k, proving necessity. Conversely, induct on the number of cosets in a pairwise intersecting family. A family of at most two already has a common element. For a larger family, the induction hypothesis gives a common element of all but one coset. Right multiplication by its inverse turns those cosets into the base cosets P_j. The remaining coset meets P_{j_0}, so it has a representative $g \in P_{j_0}$. Its intersections with the other base cosets are precisely the product conditions in the statement. The criterion supplies k, and undoing the right translation gives a common element of the original family. □

Lemma C.HellyCount (finite recognition). For a finite group, put

$$X_{ij} = \{P_i x : x \in P_j\}, \qquad P_J = \bigcap_{j\in J} P_j.$$

Typed coset Helly is equivalent to

$$\left|\bigcap_{j\in J} X_{ij}\right| = [P_J : P_J \cap P_i] \quad (i \notin J,\ |J| \geq 2). \tag{C3}$$

Proof. The cosets represented in P_J form a subset of the displayed intersection, of size the right side. Equality is therefore equality of these sets. This is exactly the anchored condition of UC.Helly: a coset meeting every base P_j has a representative in their common subgroup. The finite count proves both directions. □

Theorem UC.Action (faithful action after retaining objects). Let G act on sets X and Y. Suppose a G-invariant subset $Y_0 \subseteq Y$ has an equivariant surjection onto X. Suppose also that Y has an injective equivariant encoding in a tagged disjoint union of finite products, each factor of which is X or its power set, and that the tags are fixed by G. Then

$$\ker(G \curvearrowright Y) = \ker(G \curvearrowright X) = \bigcap_{x\in X} G_x. \tag{5.1}$$

If X is a disjoint union of transitive orbits with selected stabilizers H_j, the common kernel is

$$N = \bigcap_j \operatorname{core}_G(H_j) = \operatorname{core}_G\left(\bigcap_j H_j\right). \tag{5.2}$$

The faithful permutation image is G/N. If its action on Y is chamber-transitive on a thin, residually connected finite-rank geometry, then G/N is the full type-preserving automorphism group. The preimages in G of a chamber stabilizer and of a panel stabilizer are respectively N and a subgroup of index two over N.

Proof. An element fixing Y pointwise fixes X through the equivariant surjection. Conversely, an element fixing X fixes all subsets of X, every finite tuple formed from these objects, and all fixed tags. Injectivity of the encoding then forces it to fix Y pointwise. This proves (5.1). In a transitive orbit all point stabilizers are conjugates of the selected one, so intersection over all points gives the first equality in (5.2). The second is the intersection of the same conjugate subgroups in the opposite order. The first isomorphism theorem identifies the faithful image.

By UC.Geometry that image has trivial chamber stabilizers and is the full type-preserving group. Its panel stabilizer acts transitively on the two chambers through the panel, with trivial stabilizer of either chamber, so has order two. Taking preimages proves the final assertion. □

Independently labeled copies of the same subset need extra data in their encoding; they cannot be silently treated as the subset itself.

Proposition UC.ExtensionOrbits (classification above a retained base). Let a group B act on sets Z, E, and let $p : Z \to E$ be equivariant. For one representative e of each orbit in $B\backslash E$, there is a natural bijection

$$B\backslash Z \;\cong\; \coprod_{e\in B\backslash E} B_e\backslash p^{-1}(e). \tag{5.6}$$

Proof. Translate any $z \in Z$ so that $p(z)$ is the selected representative of its orbit. If two translated points over that representative are B-equivalent, an element taking one to the other fixes their image e, so belongs to B_e. Conversely a B_e-equivalence is already a B-equivalence. These two observations prove surjectivity and injectivity of (5.6). □

3 Endpoint conjugation and central twists

Let $(G; a, b, (r_i)_{i\in E})$ be a C-group marking with E finite and $[a, r_i] = 1$ for all $i \in E$. No string diagram, finite group, or underlying geometry is assumed. Define

$$\begin{gathered} a' = aba, \quad R_J = \langle r_j : j \in J\rangle, \quad U_J = \langle a, b, R_J\rangle, \quad H_J = \langle a', b, R_J\rangle, \\ P = \langle b, R_E\rangle, \quad H = H_E, \quad \mathcal{F} = \{J \subseteq E : a \in H_J\}. \end{gathered} \tag{E1}$$

Conjugation by a exchanges b and a' and fixes R_J pointwise. Consequently

$$H_J \trianglelefteq U_J = H_J\langle a\rangle, \qquad [U_J : H_J] \le 2. \tag{E2}$$

Theorem C.Endpoint. The new marking $(H; a', b, (r_i))$ has IP if and only if

$$P \cap P^a = R_E, \qquad \mathcal{F} \text{ is closed under intersections.} \tag{E3}$$

The group H is normal of index at most two in G. It is G precisely when $\mathcal{F}$ is nonempty. Since E is finite, a nonempty intersection-closed $\mathcal{F}$ is precisely the family of supersets of one least support $S \subseteq E$.

Proof. The family $\mathcal{F}$ is upward closed. Any new marked subgroup is of one of the forms

$$A_{\epsilon,J} = \langle b^\epsilon, R_J\rangle\ (\epsilon = 0, 1), \qquad H_J, \qquad Q_J^a \text{ with } Q_J = \langle b, R_J\rangle, \tag{E4}$$

where b^0 means that b is omitted. These correspond to omitting a', using both endpoint labels, or using a' but omitting b. Intersections of two A's are old marked intersections. Also

$$A_{\epsilon,I} \cap H_J \le A_{\epsilon,I} \cap U_J = A_{\epsilon,I\cap J};$$

all generators on the right belong to H_J, so equality holds. Intersections of two Q^a's follow by conjugating old IP. Conjugation by a normalizes H_I, so an H-with-Q^a intersection reduces to the just proved H-with-A case. For an A-with-Q^a intersection, the first condition in (E3) confines it to R_E. Intersecting there and using that a centralizes R_E gives $R_{I\cap J}$.

It remains to intersect H_I and H_J. Put $K = I \cap J$. Old IP gives

$$H_K \le H_I \cap H_J \le U_I \cap U_J = U_K. \tag{E5}$$

If $a \in H_K$ then $H_K = U_K$. Otherwise H_K has index two in U_K, so the middle subgroup is either H_K or U_K. The latter case occurs exactly when a belongs to both H_I and H_J, and intersection closure excludes it. This proves all new marked intersections. Necessity of the first condition follows by intersecting the two endpoint-deleted new parabolics; necessity of the second follows by intersecting H_I, H_J. The generators are involutions and the proved IP supplies independence. Equation (E2) with $J = E$ gives the group and index assertions. A finite nonempty upward intersection-closed family contains its total intersection S, giving the least-support description. □

Corollary C.Odd (odd endpoint and proper halving). If $m = |ab|$ is odd, the only test in (E3) is the overlap $P \cap P^a = R_E$, and $H = G$. If H is proper in G, the only test is again the overlap, and $[G : H] = 2$. In either case the endpoint product has order m/gcd(m,2) for finite m; infinite m remains infinite.

Indeed $a'b = (ab)^2$, and for odd m,

$$a = (a'b)^{(m+1)/2}b \in H_\emptyset.$$

Thus $\mathcal{F}$ is respectively all subsets or empty. An even endpoint order by itself does not imply that H is proper; the least-support alternative in Theorem C.Endpoint cannot be discarded.

Lemma C.Twist (central twist). Let a generating family of a group G contain involutions b,c with $|bc| = m$ odd and $m \geq 3$. Suppose $w \in G$ is a nonidentity involution centralizing b and c. Replacing b by wb in that family preserves G and changes $|bc|$ to $2m$.

Proof. The commuting factors w and bc have coprime orders, so $(wbc)^m = w$ and $|wbc| = 2m$. Hence the new generators recover w and b. The new marker wb is a nonidentity involution: $wb = 1$ would make $b = w$ commute with c, contrary to $m \geq 3$ odd. □

A parabolic containing w,b,c is unchanged by this twist; one omitting b is unchanged. Other intersections and typed Helly still require proof. Lemma C.Twist is the auxiliary operation needed in the finite application below, not a blanket geometric preservation theorem.

4 Graph halving and lattice polytopes

Let Γ be a finite-rank residually connected incidence geometry, of rank $n \geq 3$. Designate types 0 and 1 as vertices and edges. Assume their truncation is a simple graph Q, and that an edge is incident with a higher object exactly when both its endpoints are incident with that object. For a flag F of higher objects, let Q_F be its common vertex-edge shadow. All Q_F are connected and contain an edge.

UC.Geometry supplies the connected chamber graph of every residually connected finite-rank geometry. Restricting a chamber gallery to any set of at least two types shows that the corresponding truncation is connected, because successive complete incidence subgraphs overlap. Apply this inside the residue at F; chamber extension supplies an edge. The edge-shadow hypothesis says Q_F is the induced graph on its common vertices.

For a connected graph Q, its bipartite double cover D(Q) has vertices (v, ε), $\varepsilon \in \{0, 1\}$, with (v, ε) adjacent to $(w, 1 - \varepsilon)$ when v is adjacent to w. It is connected when Q has an odd cycle, and has two components otherwise: parity of walks proves both assertions. Choose a component C of D(Q). Define $L_C(\Gamma)$ as follows. Its first two types are (v,0) and (v,1) in C. A higher object is a pair (x, C_x), where C_x is a component of $D(Q_x)$ contained in C. Incidence of the first two types is adjacency, incidence with a higher object is membership in C_x, and two higher objects are incident when their old objects are incident and their chosen components meet.

The precise compatibility requirement is

$$\text{if every } Q_x \ (x \in F) \text{ is nonbipartite, then } Q_F \text{ is nonbipartite,} \tag{PC}$$

for every nonempty higher flag F. It is automatic if the higher shadows of every flag are nested, or if Q is bipartite.

Theorem C.Graph. The system $L_C(\Gamma)$ is always a geometry. It is residually connected exactly when (PC) holds. It is thin exactly when Γ is thin. A chamber-transitive automorphism group of Γ acts faithfully and chamber-transitively on $L_C(\Gamma)$ when Q is nonbipartite; when Q is bipartite its index-two color-preserving subgroup does so.

Proof. For a flag $\Phi = \{(x, C_x) : x \in F\}$ of higher lifts, write $W = \cap C_x$. If any Q_x is bipartite, restriction of C_x to $D(Q_F)$ is one component. Any two such restrictions agree: incidence makes them meet over the connected $Q_{x,y}$, and hence select the same parity class there and over Q_F. Nonbipartite

constituents impose no restriction. Thus W is one component if any constituent is bipartite, and all of $D(Q_F)$ otherwise. In particular W always contains an edge, and is connected exactly under (PC) for this flag. For F empty use $W = C$.

An edge of W and a chamber of Γ extending its base edge and F give a chamber extending Φ. Every additional higher object has a unique component containing the chosen lift of that edge. The same extension works for a flag already containing one lifted vertex. If it contains both, their base edge is unique by simplicity and incident with all the old higher objects by the shadow condition. This proves the geometry axiom.

The residue at (v,0) is the old residue Γ_v: a neighbour (w,1) corresponds to the unique edge {v,w}, and the component for an incident higher object is uniquely specified by (v,0). Simplicity and the shadow condition preserve all incidences. The other color is identical. Hence residues of flags containing a lifted vertex are connected. For a flag of higher objects only, its two-type truncation is W; when W is connected, every other residual object attaches to it through a chamber. This proves residual connectivity under (PC). If (PC) fails, choose its higher flag; all constituents have unique lifts but W has two components. Were the resulting geometry residually connected, the truncation lemma inside its residue would make W connected. This proves necessity without treating a truncation as a residue.

Panels containing both new vertex types correspond to old panels missing the same higher type. A panel missing one new vertex type corresponds to an old panel missing the edge type, at the retained endpoint. Old panels missing the vertex type already have two objects by simplicity. These correspondences prove both directions of thinness.

If Q is nonbipartite, an old chamber $(v, \{v, w\}, x_2, \dots, x_{n-1})$ corresponds bijectively to the new chamber with (v,0),(w,1) and the components containing them. Old automorphisms act naturally on this bijection. If Q is bipartite, the same bijection uses the old chambers whose chosen vertex is in one fixed color class. A chamber-transitive group contains a color-exchanging element, since it exchanges the endpoints of a base edge through a chamber move. Its color-preserving subgroup therefore has index two and is transitive on the selected chambers, with the same chamber bijection. Faithfulness in both cases follows from the retained-object argument below. □

4.1 The retained objects

For completeness, apply UC.Action to the precise object sets in this construction. Let K be G when Q is nonbipartite and its color-preserving subgroup otherwise. Take X to be the disjoint union of old vertices and higher objects, and Y the object set of the halved geometry. The map

$$(v, \varepsilon) \mapsto v, \qquad (x, C_x) \mapsto x \tag{C4}$$

is an equivariant surjection $Y \to X$. For nonbipartite Q the chosen C is all D(Q); for bipartite Q it contains exactly one lift of every old vertex. In the latter case its intersection with the double cover of each connected shadow Q_x is one of that double cover's two components. Thus every higher object also has a lift.

Encode a new vertex by its fixed color tag and old vertex. Encode (x, C_x) by the old object x and the two subsets $\{v : (v, 0) \in C_x\}, \{v : (v, 1) \in C_x\}$ of X, retaining its old type as a tag. Components are determined by their vertex sets, so the encoding is injective and equivariant. UC.Action gives

$$\ker(K \curvearrowright Y) = \ker(K \curvearrowright X) = 1. \tag{C5}$$

The last equality follows because an old edge is uniquely determined by its two endpoints in the simple graph Q. Fixing X therefore fixes all old objects. When Γ is thin and (PC) holds, the proved geometry is thin and residually connected, and its K-action is chamber-transitive. UC.Geometry then identifies K with the full type-preserving group. This argument does not infer the geometry or its flag properties merely from faithfulness.

Suppose now Γ is regular and thin, with vertex-changing marker a, edge-changing marker b, and higher markers r_i; assume a commutes with the higher markers. Under (PC), Theorem C.Graph produces a regular thin residually connected geometry. Directly on a base edge, its panel involutions are b^a, b, r_i: b changes one endpoint, aba changes the other, and each r_i changes its higher object. UC.Geometry identifies their group as the full type-preserving group H. Lemma C.Gallery identifies the stabilizer of every base subflag with the subgroup generated by its missing-type markers, giving IP and identifying the actual coset incidence with $L_C(\Gamma)$.

There is also an exact correspondence between the two parity conditions. In the base residue with group U_J from (E1), the vertex stabilizer is $Q_J = \langle b, R_J \rangle$. Its vertex graph is bipartite precisely when $a \notin H_J$. If H_J is proper, its two cosets give a coloring, since $Q_J \leq H_J$ and a crosses colors. Conversely the action on the two colors supplies a character killing b and all r_j and sending a to the nontrivial element, so it excludes a from H_J. In a globally nonbipartite geometry, higher flags reduce by transitivity to base flags with shadows $U_{E\setminus S}$. Condition (PC) says

$$[E \setminus \{i\} \in \mathcal{F} \text{ for all } i \in S] \Longrightarrow E \setminus S \in \mathcal{F}. \tag{G1}$$

An upward family containing E satisfies (G1) exactly when it is principal: let T be the set of i whose complementary coatom belongs to $\mathcal{F}$; (G1) gives $E \setminus T \in \mathcal{F}$, and every member of $\mathcal{F}$ contains $E \setminus T$. Thus geometric parity compatibility is the least-support condition of Theorem C.Endpoint. The mixed overlap in that theorem has a different role: it says the two adjacent vertices determine one edge. Indeed P and P^a stabilize the endpoints; their intersection is the edge's pointwise stabilizer R_E exactly when no other edge has those endpoints, using transitivity on panels and the regular chamber identification.

4.2 Halving lattice polytopes

Corollary C.Halving. Let P be a regular rank-n abstract polytope, $n \geq 3$, whose face poset is a lattice. Replacing one endpoint marker a by aba gives a C-group H normal of index at most two in Aut(P). Its maximal-parabolic coset system is a regular hypertope, with full type-preserving group H. The diagram duplicates the edge labels from b to the remaining markers, and the a'–b label is m/gcd(m,2) for $m = |ab|$ finite, or infinity for infinite m. The index is two exactly when the corresponding vertex graph is bipartite; for a final-end operation use the dual graph on facets.

Proof. In a lattice polytope an edge is the join of its two distinct vertices. Thus two vertices determine at most one edge, and that edge belongs to a higher face exactly when both vertices do. Higher faces of a flag form a chain, so their shadows are nested and (PC) is automatic. Theorem C.Graph and the panel-marker identification prove the asserted geometry, full group, and IP without an alternating-semiregular-polytope theorem. Equation (E2) and the coloring argument give the index. Finally $a'b = (ab)^2$ and a commutes with all r_i, so $a'r_i$ is conjugate to br_i; all other products are unchanged. □

5 Graph differences and even subsets

For a nonempty set Ω, let $\mathbb{F}_2^{(\Omega)}$ be the vector space of finitely supported functions on Ω, with coordinate vectors e_v. Its augmentation module is

$$E(\Omega) = \left\{ z \in \mathbb{F}_2^{(\Omega)} \,:\, \sum_{v \in \Omega} z_v = 0 \right\}.$$

For a chosen vertex o, the vectors $e_o + e_v$, $v \neq o$, are a basis. Indeed a vector of even total sum is the sum of $e_o + e_v$ over its nonzero coordinates away from o; independence follows by inspecting those coordinates. Thus, if $|\Omega| = m < \infty$, the module has order 2^{m-1}.

Theorem 1 (Presentation by graph differences)**.** *Let Q be a connected simple graph with vertex set Ω. Define $B(Q)$ by one generator z_e for every edge, with relations*

$$z_e^2 = 1, \qquad [z_e, z_f] = 1, \qquad z_{e_1} \cdots z_{e_k} = 1$$

for every finite closed walk with successive edges $e_1, \dots, e_k$. Then $z_{\{u,v\}} \mapsto e_u + e_v$ defines an isomorphism $B(Q) \cong E(\Omega)$. It commutes with every graph automorphism.

Proof. The edge vectors have order dividing two and commute. Their sum along a closed walk is zero because each successive vertex occurs twice. Hence the indicated homomorphism exists.

Choose a base vertex o. For each v, choose a finite path from o to v and let t_v be the product of its edge generators. The product is independent of the chosen path: concatenate one path with the reverse of the other, and apply the closed-walk relation. In particular $t_o = 1$. Following a path to u, then the edge $\{u, v\}$, and finally a reverse path from v to o gives $z_{\{u,v\}} = t_u t_v$.

The elements t_v commute and have square one. The basis description above therefore defines a homomorphism $E(\Omega) \to B(Q)$ sending $e_o + e_v$ to t_v. The two composites fix every edge generator and every basis vector, respectively, so they are identity maps. The original edge map, unlike the chosen inverse formula, uses no base vertex or paths. It consequently commutes with every graph automorphism. □

For a finite graph one may use a spanning tree and only the fundamental cycles determined by the remaining edges. The same proof applies: the unique tree paths define t_v, and each fundamental-cycle relation gives $z_{\{u,v\}} = t_u t_v$ for its extra edge. This also proves directly that every closed-walk relation follows from that finite list.

Corollary 2 (Generation inside a permutation extension)**.** *Let H act by graph automorphisms on a connected graph Q. In $E(\Omega) \rtimes H$, suppose a subgroup contains $\{0\} \rtimes H$ and translations $e_u + e_v$ for a set of edges whose H-orbits form a connected spanning subgraph. Then it is the whole semidirect product.*

Proof. Conjugation by $(0, h)$ carries the translation $e_u + e_v$ to $e_{hu} + e_{hv}$. The subgroup therefore contains every edge difference in the indicated spanning subgraph. Theorem 1 says that these generate $E(\Omega)$, proving the assertion. □

6 Recovering a semidirect product from its translations

Let H act on a nonempty set Ω, and fix $o \in \Omega$. We use left actions and multiplication

$$(v, g)(w, h) = (v + gw, gh) \quad \text{in } E(\Omega) \rtimes H.$$

The following statement does not require either set or group to be finite.

Theorem 3 (Recognition by differences from a base vertex)**.** *Let P be a group, let $j : H \to P$ be a homomorphism, and let $(t_v)_{v \in \Omega}$ be commuting elements of P satisfying $t_v^2 = 1$ and $t_o = 1$. Suppose*

$$j(h) t_v j(h)^{-1} = t_{ho} t_{hv} \qquad (h \in H,\ v \in \Omega). \tag{1}$$

There is a unique homomorphism

$$\Psi : E(\Omega) \rtimes H \longrightarrow P, \qquad \Psi(e_o + e_v, 1) = t_v, \qquad \Psi(0, h) = j(h).$$

It is surjective if these elements generate P. Under that generation hypothesis, if a homomorphism $\Phi : P \to E(\Omega) \rtimes H$ has the displayed inverse values on these elements, then Φ and Ψ are mutually inverse.

Proof. The commuting involutions define a homomorphism $T : E(\Omega) \to P$ by $T(e_o + e_v) = t_v$ on the stated basis. The action of h on this vector is

$$h(e_o + e_v) = (e_o + e_{ho}) + (e_o + e_{hv}).$$

Consequently (1) implies $T(hw) = j(h)T(w)j(h)^{-1}$ first on every basis vector and then on every vector. Define $\Psi(w, h) = T(w)j(h)$. For two such pairs,

$$T(v)j(g)T(w)j(h) = T(v)T(gw)j(gh) = T(v + gw)j(gh),$$

which is exactly the required homomorphism identity. Uniqueness and surjectivity follow from the generating sets. Under the final hypothesis each composite fixes its respective generating set, and is therefore the identity. □

Lemma 4 (A twist exchanging two complements)**.** *The function $c(h) = e_o + he_o$ satisfies $c(gh) = c(g) + gc(h)$. Thus*

$$\delta(v, h) = (v + c(h), h)$$

is an involutory automorphism of $E(\Omega) \rtimes H$. The standard complement $A = \{(0, h) : h \in H\}$ and its image satisfy

$$A \cap \delta(A) = \{(0, h) : h \in H_o\},$$

where H_o is the stabilizer of o.

Proof. In characteristic two, $e_o + ghe_o = (e_o + ge_o) + g(e_o + he_o)$, proving the cocycle identity. Substitution into the multiplication rule proves that δ is a homomorphism; applying it twice adds $c(h)$ twice, so gives the identity. Finally $(0, h)$ belongs to its image of A exactly when $c(h) = 0$, or equivalently $ho = o$. □

7 Cyclic displacement relations

The next identity is the displacement argument in Li–Zhang's proof of their Theorem 3.4. Keeping integer indices exposes the action needed in Theorem 3 and avoids a separate order argument.

Lemma 5. *Let x, y belong to a group and put $t_k = y^k x^k$ for $k \in \mathbb{Z}$. If every t_k has square one, then all t_k commute and*

$$x^{-1} t_k x = t_1 t_{k+1}, \qquad x t_k x^{-1} = t_{-1} t_{k-1}.$$

If an involution b inverts x and y, then $b t_k b = t_{-k}$. If $x^n = y^n = 1$, it suffices to require $t_k^2 = 1$ for $1 \le k < n$.

Proof. From $t_k = t_k^{-1}$ we obtain $y^k x^k = x^{-k} y^{-k}$ and, on conjugating by x^k, $x^k y^k = y^{-k} x^{-k}$. Hence

$$\begin{aligned} t_i t_j &= x^{-i} y^{j-i} x^j = x^{-j} (x^{j-i} y^{j-i}) x^j \\ &= x^{-j} y^{i-j} x^i = t_j t_i. \end{aligned}$$

Direct multiplication gives $x^{-1} t_k x = t_1^{-1} t_{k+1} = t_1 t_{k+1}$. For $k = -1$ this yields $x t_1 x^{-1} = t_{-1}$. Conjugate the first formula with index $k - 1$ by x and use this identity to obtain the second formula. Conjugation by b gives $b t_k b = y^{-k} x^{-k} = t_{-k}$. Finally the two period relations give $t_{k+n} = t_k$ for every integer k, while $t_0 = 1$; this proves the finite-relation assertion. □

Corollary 6. *Suppose $n \ge 3$ and $P = \langle x, b, y \rangle$ satisfies*

$$x^n = y^n = b^2 = 1, \quad bxb = x^{-1}, \quad byb = y^{-1}, \quad (y^k x^k)^2 = 1 \quad (1 \le k < n).$$

Let $H = D_{2n}$ act on $\mathbb{Z}/n\mathbb{Z}$ by $a(i) = i + 1$ and $b(i) = -i$. If there is a homomorphism $\Phi : P \to E(\mathbb{Z}/n\mathbb{Z}) \rtimes H$ with

$$x \longmapsto (0, a), \quad b \longmapsto (0, b), \quad y \longmapsto (e_0 + e_{-1}, a^{-1}),$$

then it is an isomorphism. Its inverse sends $(e_0 + e_i, 1)$ to t_{-i} and $(0, a), (0, b)$ to x, b.

Proof. The dihedral relations define $j : H \to P$, with $j(a) = x$ and $j(b) = b$. Set $T_i = t_{-i}$; the period relations make this independent of the integer representative of i. Lemma 5 gives commuting involutions and

$$xT_ix^{-1} = T_1T_{i+1}, \qquad bT_ib = T_{-i}.$$

These are exactly (1) for a, b, with $o = 0$. Equivariance for products and inverses follows by applying these identities successively; thus it holds for all $h \in H$. The elements $j(H), T_i$ generate P, since $y = t_1x^{-1}$. Moreover $\Phi(t_k) = (e_0 + e_{-k}, 1)$ by direct multiplication, or by the cocycle identity in Lemma 4. The final assertion of Theorem 3 now gives the inverse and the isomorphism. □

8 Formal components and their limits

The file `SemidirectRecognition.lean` uses mathlib's standard semidirect-product construction. It verifies the actual lift $\Psi(n, h) = T(n)j(h)$, proves its surjectivity from the stated generation hypothesis, and constructs the multiplicative equivalence from the two component identities. Its nine reported declarations make no finiteness assumption. The augmentation basis and the passage from cyclic displacement generators to these component homomorphisms are written arguments above, rather than implicit formal hypotheses.

Two unchanged files are also included: the five reported displacement declarations in `BinaryDihedralDisplacements.lean`, and the two mixed-intersection declarations in `EndpointDoublingIntersections.lean`. The latter assumes the fixed-overlap hypotheses of the appropriate case of C.Endpoint. These components do not formalize the complete endpoint criterion, coset geometry, graph halving, or source-specific applications. The separate checker recompiles all three files and reports these limits explicitly.

References

[MW20] A. Montero and A. Ivić Weiss, *Locally spherical hypertopes from generalised cubes*, arXiv:2001.11082v1, 29 January 2020.

[LZ25] M. Li and W.-J. Zhang, *Regular 3-polytopes of type $\{n, n\}$*, arXiv:2505.09505v1, 14 May 2025.

[PT24] C. A. Piedade and P. Tranchida, *Constructing new geometries: a generalized approach to halving for hypertopes*, arXiv:2405.19050v1. The graph construction and lattice convention are used at the explicit scope stated above.